Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/01/2012
PI Name:	Bassham, Diane Ph.D.		
Project Title:	Molecular Basis for the Role of the Vacuole in Shoot Gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bassham@iastate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	515-294-7461
Organization Name:	Iowa State University		
PI Address 1:	Genetics, Development and Cell Biology		
PI Address 2:	1035B Carver Co-Lab		
PI Web Page:			
City:	Ames	**State:**	IA
Zip Code:	50011-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	08/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2013, per NSSC information (Ed. 5/8/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AK78G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant perception and response to gravity is critical for appropriate growth, development, and environmental responses. Shoot gravitropism involves the perception of gravity by sedimentation of starch-containing amyloplasts in gravity-sensing cells, followed by transmission of the signal to additional cell types to promote directional growth. Vesicle trafficking to the vacuole is essential for the perception of the gravity signal, but the molecular basis for this requirement is unknown. Our long term goal is to determine the mechanism by which gravity is perceived by shoots and the gravity signal transduced to other cell types to produce a response. The objective of this application is to determine the function of vacuolar trafficking proteins in shoot gravitropism. We hypothesize that appropriate physical properties of the vacuole are essential for amyloplast sedimentation, and therefore for gravity perception. We will test this hypothesis by measuring the elasticity of wild type and mutant vacuoles using atomic force microscopy and by comparing the lipid constituents of the vacuolar membranes. We will also identify additional proteins involved in vacuolar trafficking by a combination of in vitro membrane fusion assays and in vivo immunoprecipitations and test their function in gravitropism using genetic approaches. This research will provide a molecular explanation for the observation that shoot gravitropism is specifically defective in vacuolar trafficking mutants and allow us to make testable predictions about the mechanistic basis for their agravitropic phenotype. It will also provide a foundation for the generation of transgenic plants with altered gravity response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary impact of our research will be to increase understanding of the mechanisms by which plants respond to gravity. In addition to being important for the growth of plants in the altered gravitational conditions likely to be encountered during space flight, this research will also contribute to our knowledge of plant growth, development, and response to environmental cues. The gravitational response interacts extensively with other developmental, hormonal, and environmental response pathways, leading to morphological and anatomical changes to the plant structure. Understanding of the molecular mechanism of gravitropism is therefore vital for the development of strategies to improve plant growth and yield under varying conditions. The development of technologies to analyze the physical properties of vacuoles is also likely to impact research in plant cell structure and stress responses.
Task Progress:	Personnel Rahul Roy (PhD student) is continuing to work full time on this project. A PhD rotation student, Xiaochen Yang, has also contributed to the project this year, along with two undergraduate students. Research progress Objective #1. Analysis of vacuolar properties in the agravitropic vti11 vacuolar trafficking mutant. We are developing a novel microfluidics platform for the analysis of vacuolar physical and structural properties. Microfluidics channels have been designed and synthesized. Initial experiments will involve changes in osmotic potential of the buffer system to determine the lysis point of vacuoles of different genotypes. Preliminary experiments using protoplasts have demonstrated the feasibility of the approach and are allowing us to optimize conditions for buffer exchange. Objective #2. Identification of additional vacuolar trafficking components that are required for shoot gravitropism, and characterization of their molecular function. We previously reported the identification and phenotypic characterization of a new vacuolar trafficking mutant, tno1. We have now analyzed in detail the phenotype of this mutant with respect to gravitropism. A statistically significant defect in gravitropic responses of etiolated hypocotyls of the mutant has been demonstrated using two different assays. Roots also appear affected in gravitropic responses, although this has not yet been quantified. By contrast, inflorescence stems do not show gravitropic defects. From these results we conclude that TNO1 is involved in the gravitropic response of hypocotyls and probably also roots but not of inflorescence stems and we are now attempting to determine the basis for the gravitropic phenotype. A new weak allele has been identified for the vacuolar trafficking component VPS45, for which a knockout mutation is lethal. This mutant has defects in both shoot and root gravitropism as well as general growth and morphological defects. A homozygous mutant in which expression of the t-SNARE SYP51 is reduced has been identified. Initial gravitropism assays show no significant shoot gravitropism defects, possibly due to functional redundancy or the residual SYP51 expression. Further analysis of each of these mutants is ongoing.
Bibliography Type:	Description: (Last Updated: 09/18/2019)
Articles in Peer-reviewed Journals	Contento AL, Bassham DC. "Structure and function of endosomes in plant cells." Journal of Cell Science. 2012. In press as of May 2012. , May-2012